

Lab: Designing Your Iteration Roadmap

Objective

Plan a structured sequence of prototype iterations for your invention project, specifying what each iteration tests, what decisions hinge on the results, and how the action-perception loop will drive convergence toward a working solution.

Prerequisites

- Familiarity with the action-perception loop, MVP concept, A/B testing, and pivot-or-persevere decisions from this module's lecture
- Your prototype specification from Module 01 and test results (real or hypothetical) from Modules 03-04
- Your current list of unresolved design questions

Materials

- Large sheet of paper or whiteboard for roadmap visualization
- Sticky notes or index cards
- Your prototype and test records from prior modules

Part 1: Identify Your Design Uncertainties (10 minutes)

Goal: Prioritize the unresolved questions that your iteration sequence must address.

List all remaining design uncertainties — things about your invention that you do not yet know. Then rank them by criticality: how much does resolving this uncertainty affect the viability of your invention?

Uncertainty	Criticality (1-5)	Current state of knowledge	What test would resolve it?

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Rank order your uncertainties from most critical to least critical. Your iteration sequence should address the most critical uncertainties first.

Write your response here

Part 2: Define Your Minimum Viable Prototype (10 minutes)

Goal: Determine the simplest prototype that tests your most critical uncertainty.

Take your #1 ranked uncertainty from Part 1. Design the minimum viable prototype that could resolve it:

Uncertainty being tested:

Write your response here

What the prototype must include to generate relevant prediction errors:

Write your response here

What the prototype can exclude (not relevant to this specific test):

Write your response here

Predicted cost to build this MVP:

Write your response here

Predicted time to build:

Write your response here

What observation would confirm the hypothesis?

Write your response here

What observation would disconfirm it?

Write your response here

Now compare: is your current prototype (or planned prototype) more complex than this MVP?
If so, what components could you remove to get to minimum viability faster?

Write your response here

Part 3: Design an A/B Comparison (10 minutes)

Goal: Practice designing a controlled comparison between two design alternatives.

Identify one design choice you are currently uncertain about — a fork in the road where you have two plausible options:

The design choice:

Write your response here

Option A:

Write your response here

Option B:

Write your response here

Design the A/B test:

Test Parameter	Details
What varies between A and B	

Write your response here

|| What stays constant (controlled variables) |

Write your response here

|| How are test conditions assigned (random, sequential, paired?) |

Write your response here

|| What do you measure? |

Write your response here

|| How many tests/users per condition? |

Write your response here

| | What result would make A the winner? |

Write your response here

| | What result would make B the winner? |

Write your response here

| | What result would be inconclusive? |

Write your response here

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Part 4: Pivot-or-Persevere Assessment (15 minutes)

Goal: Honestly evaluate whether your current approach warrants continued refinement or a structural change.

This exercise requires candor. Answer the following diagnostic questions:

Iteration trajectory: - How many iterations have you completed?

Write your response here

- Is each iteration producing measurable improvement?

Write your response here

- Are the improvements getting larger, staying constant, or getting smaller (diminishing returns)?

Write your response here

Prediction error trends: - Are your prototype's prediction errors decreasing with each iteration?

Write your response here

- Are the remaining errors parameter-level (tuning) or structural (fundamental approach)?

Write your response here

External signals: - Does user/expert feedback suggest the basic approach is right but needs refinement, or that the approach itself is wrong?

Write your response here

- Has any competitor or alternative approach solved the same problem differently? If so, does their approach generate lower prediction errors?

Write your response here

Pivot-or-persevere assessment:

Signal	Your Situation	Interpretation
Iterations producing diminishing returns		

Write your response here

| Persevere / Lean toward pivot | | Users describe the core concept as valuable |

Write your response here

| Persevere / Lean toward pivot | | Prediction errors are structural, not parametric |

Write your response here

| Persevere / Lean toward pivot | | A specific structural change could open new design space |

Write your response here

| Persevere / Lean toward pivot |

Your recommendation to yourself: Persevere, pivot, or gather more data? Justify in 2-3 sentences.

Write your response here

Part 5: Build the Iteration Roadmap (15 minutes)

Goal: Create a visual roadmap of your next 3-5 prototype iterations with decision gates.

Fill in the roadmap:

Iteration	Hypothesis Being Tested	Fidelity Level	Estimated Time/ Cost	Success Criteria	Failure Criteria	Next step if success	Next step if failure
v1 (MVP)							

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|| v2 |

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Decision gates: - After which iteration will you make the major persevere/pivot decision?

Write your response here

- What evidence would trigger an early pivot before reaching that gate?

Write your response here

Discussion and Debrief

1. **MVP discipline:** Was it difficult to strip your prototype down to the minimum viable version? What psychological pull did you feel to include "just one more feature"?
2. **A/B clarity:** Did designing the A/B test force you to clarify what you are actually uncertain about? Was the design choice harder or easier to define than you expected?
3. **Pivot honesty:** Were you able to honestly assess whether your project shows persevere or pivot signals? What made this assessment emotionally challenging?
4. **Roadmap realism:** Does your iteration roadmap feel achievable? Where are the resource bottlenecks? What could you do to speed up the loop frequency?

5. **Loop speed:** If you could run the action-perception loop twice as fast (half the iteration time), how would that change your strategy? What would you sacrifice for speed?

Write your response here